BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, RAWALPINDI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Mathematics (Objective) — (Group-I) — HSSC-(P-II)-A/2023 (For All Sessions) — Paper Code: 8191

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 8191

Note: Note: Write Answers to the Questions on the objective answer sheet provided. Four possible answers A, B, C and D to each question are given. Which answer you consider correct, fill the corresponding circle A, B, C or D given in front of each question with Marker or Pen ink on the answer sheet provided.

Q1. $\lim_{x\to 0} (e^{2x} - 1)/x = ?$ [numbered 1.1 as printed]

(A) 1
(B) 2
(C) 0
(D) -2

Q2. coshx + sinhx = ?

(A) e^x
(B) e^{-x}
(C) e^{2x}
(D) $2e^x$

Q3. $d/dx[\ln(2^x)] = ?$

(A) ln2
(B) 2^x
(C) $1/2^x$
(D) $\ln 2/2^x$

Q4. d/dx(coshx) = ?

(A) -sinhx
(B) sinhx
(C) sechx
(D) cosechx [sic]

Q5. If $f(x) = \sqrt{x}$, then $f'(0) = ?$

(A) 0
(B) 1
(C) Undefined
(D) 1/2

Q6. $d/dx(\sin^{-1}x + \cos^{-1}x) = ?$

(A) 1
(B) 0
(C) -1
(D) 2

Q7. $\int 0\, dx = ?$

(A) x
(B) 1/x
(C) x^2
(D) C

Q8. $\int e^x(x + 1)\, dx = ?$

(A) $xe^x + c$
(B) $e^x + c$
(C) [illegible in source scan — covered]
(D) [illegible in source scan — covered]

Q9. $\int_0$^(π/2) cosx dx = ?

(A) 0
(B) -1
(C) 1
(D) 2

Q10. $\int (\sin 2x)/(\sin x)\, dx = ?$

(A) 2 cosx + c
(B) 2 sinx + c
(C) (1/2) sinx + c
(D) (1/2) cosx + c

Q11. The slope of a line x = 5 is:

(A) 0
(B) 1
(C) -1
(D) Infinite

Q12. Midpoint of (0, -2) and (-2, 0) is:

(A) (0, 0)
(B) (-1, -1)
(C) (-2, -2)
(D) (0, -1)

Q13. Distance between (-1, 2) & (7, 5) is:

(A) √73
(B) 7
(C) 2√73
(D) 73

Q14. The solution of inequality $x + 2y < 6$ is: [best read — partially covered]

(A) (1, 4)
(B) (1, 3)
(C) (1, 1)
(D) (1, 5)

Q15. Equation of Tangent to $x^2 + y^2 = 4$ at (2, 0) is:

(A) x = 1
(B) y = 1
(C) y = 2
(D) x = 2

Q16. Slope of tangent to parabola $y^2 = 4ax$ at (a, 2a) is:

(A) 2 (B) -1

(C) 1 (D) 3

Q17. Eccentricity e of a circle is:

(A) $e = 0$ (B) $e = 1$

(C) $0 < e < 1$ (D) $e > 1$

Q18. Radius of a circle $x^2 + y^2 = 2$ is:

(A) 2 (B) 1

(C) 1/2 (D) $\sqrt{2}$

Q19. If P = (2, 3), Q = (6, -2), then $|\vec{PQ}|$ is

(A) $\sqrt{40}$ (B) $\sqrt{42}$

(C) $\sqrt{41}$ (D) $\sqrt{43}$

Q20. For a vector $V = 2i + 3j - 6k$, then $\cos\beta = ?$

(A) 3/7 (B) 2/7

(C) -6/7 (D) 3/7

Mathematics (Subjective) — (GROUP-I) — HSSC-(P-II)-A/2023 (For All Sessions)

Time Allowed: 2:30 hours | **Maximum Marks:** 80

Q2. Write short answers of any eight parts from the following: (8x2=16)

(i) Express perimeter P of a square as a function of its area A.

(ii) Evaluate $\lim_{x\to 0} (\sin x^0)/x$ [sic]

(iii) Define even function with example.

(iv) If $y = x^4 + 2x^2 + 2$, prove that $dy/dx = 4x\sqrt{(y-1)}$

(v) Find derivative by definition $1/\sqrt{x}$

(vi) Find dy/dx, $xy + y^2 = 2$

(vii) Differentiate w.r.t x, $y = x^2\sec 4x$

(viii) Differentiate w.r.t x, $y = \cot^{-1}(x/a)$

(ix) Find dy/dx if $y = x\sqrt{(\ln x)}$ [best read — partially covered]

(x) Apply the Maclaurin series to prove that: $e^{2x} = 1 + 2x + (4x^2)/2! + (8x^3)/3! + \ldots$

(xi) Graph the solution set of $2x + y \leq 6$.

(xii) Define feasible region.

Q3. Write short answers of any eight parts from the following: (8x2=16)

(i) Evaluate $\int \tan^2 x \, dx$.

(ii) Evaluate $\int ((a-b)x)/((x-a)(x-b)) \, dx$

(iii) Evaluate $\int x \sin x \, dx$.

(iv) Evaluate $\int_{\pi/6}^{\pi/3} \cos t \, dt$

(v) Solve the differential equation $y \, dx + x \, dy = 0$

(vi) Evaluate $\int x^2/(4+x^2) \, dx$

(vii) Find the areas between the x-axis and the curve $y = x^2 + 1$ from $x = 1$ to $x = 2$

(viii) Find a unit vector in the direction of $V = (1/2)i + (\sqrt{3}/2)j$

(ix) Find direction cosines of $V = 4i - 5j$ [best read — partially covered]

(x) Find α, so that vector $u = 2\alpha i + j - k$, $v = 1+\alpha j + 4k$ are perpendicular.

(xi) Find the area of parallelogram whose vertices are: A(0,0,0) B(1,2,3) C(2,-1,1) D(3,1,4)

(xii) A force $F = 7i + 4j - 3k$ is applied at p(1,-2,3). Find its amount about the point Q(2,1,1) [sic]

Q4. Write short answers of any nine parts from the following: (9x2=18)

(i) Is $(\sqrt{176}, 7)$ at a distance of 15 units from the origin?

(ii) By means of slopes, show that the print [sic] (-4,6), (3,8), (10,10) lie on the same line.

(iii) Find K so that the line joining A(7,3); B(k,-6) and the line joining C(-4,5), D(-6,4) are parallel.

(iv) Find the equation of the line having y-intercept -7 and slope -5.

(v) Find the point of intersection of the lines $x - 2y + 1 = 0$ and $2x - y + 2 = 0$

(vi) Find equation of lines represented by $2x^2 + 3xy - 5y^2 = 0$

(vii) Find the measure of the angle between the lines represented by $9x^2 + 24xy + 16y^2 = 0$

(viii) Find an equation of the circle with ends of diameter at (-3,2) and (5,-6)

(ix) Show that the line $2x + 3y - 13 = 0$ is tangent to the circle $x^2 + y^2 + 6x - 4y = 0$

(x) Check the position of the point (5,6) with respect to the circle $x^2 + y^2 = 81$.

(xi) Find focus and directrix of the parabola $x^2 = -16y$

(xii) Find an equation of ellipse if foci $(-3\sqrt{3}, 0)$ and vertices $(\pm 6, 0)$.

(xiii) Find equation of hyperbola with given data foci (0, ±9), directrices y = ±4

Q. SECTION-II

Note Attempt any three questions. Each question carries equal marks: (10x3=30)

5. (a) Evaluate: $\lim_{x\to 0} (\sec x - \cos x)/x$

5. (b) If $y = \tan(2 \tan^{-1} x/2)$, then show that $dy/dx = 4((1+y^2)/(4+x^2))$

6. (a) Evaluate: $\int dx / ((1/2) \sin x + (\sqrt{3}/2) \cos x)$ [best read — denominator as printed]

6. (b) Find equation of line through intersection of $x + 2y + 3 = 0$, $3x + 4y + 7 = 0$ and making equal intercepts on the axes.

7. (a) Find the area bounded by the curve $f(x) = x^3 - 2x^2 + 1$ and x-axis in the 1st quadrant.

7. (b) Minimize $Z = 3x + y$ subject to the constraints $3x + 5y \geq 15$, $x + 6y \geq 9$, $x \geq 0$, $y \geq 0$

8. (a) If $y = a \cos(\ln x) + b \sin(\ln x)$ prove that $x^2 (d2y/dx^2) + x (dy/dx) + y = 0$

8. (b) Find the coordinates of the points of intersection of the line $2x + y = 5$ and the circle $x^2 + y^2 + 2x - 9 = 0$, also find the length of intercepted chord.

9. (a) Find the centre foci, accentricity [sic] and vertices of the ellipse $x^2 + 16x + 4y^2 - 16y + 76 = 0$

9. (b) [illegible in source scan — cut at bottom edge]